

THE STRUCTURE OF MATHEMATICAL REALITY

Phase II — Chapter 6

The Unreasonable Effectiveness Problem

Quiz

Self-Study Curriculum

Before you begin

This quiz tests application, not memorisation. Answer in your own words.

Do not quote the lecture back at yourself. If you find yourself writing a phrase you remember reading, stop and rephrase it from your own understanding. A short, precise answer in your own language is worth more than a long transcription.

1. A colleague says: “Math works because the universe is mathematical.” Is this an explanation or a restatement of the mystery?

2. What is the difference between “the universe has structure” and “the universe has compressible structure”? Why does the distinction matter?

3. You discover that a biological system you are studying obeys the same equations as a completely unrelated physical system. List three possible explanations, ordered from “least surprising” to “most surprising.”

4. If mathematical effectiveness in physics is partly explained by the fact that we develop math in response to physical problems, what would constitute a genuinely “unreasonable” instance of effectiveness?

After the quiz

For each answer, ask yourself: “*Did I explain the **why**, or did I just state the **what**?*” If you stated a fact without explanation, go back and add the reasoning.

Date: _____ **Questions where I want to go deeper:**
